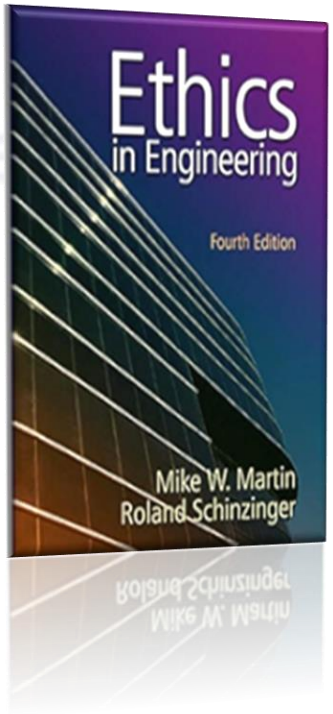
Engineering Ethics (HSS-422) (3+0)

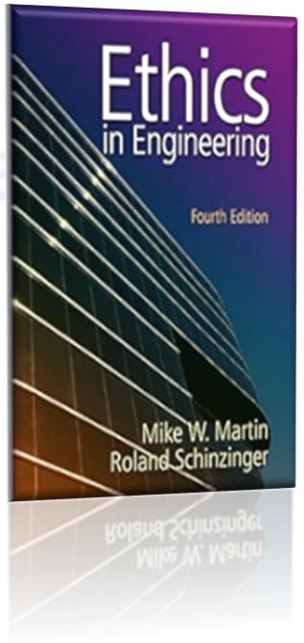






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CHALLENGER DISASTER AND TECHNOLOGY TRANSFER

Week No 12

By Engr. Syed Rizwan Ali

Learning Outcomes

In this Lecture will cover ..

- To Understand Key Ethical Principles: Gain a comprehensive understanding of the fundamental ethical principles in engineering, including public safety, integrity, honesty, and professional responsibility.
- To Analyze the Challenger Disaster: Develop the ability to critically analyze the Challenger Disaster case study, identifying the ethical dilemmas faced by engineers and the -

Learning Outcomes

In this Lecture will cover ..

- To Analyze the Challenger Disaster: dilemmas faced by engineers and the consequences of ethical failures.
- To Evaluate Technology Transfer: Learn to evaluate the ethical considerations involved in technology transfer, such as intellectual property rights, equity, fairness, and the importance of cultural sensitivity and capacity building.

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Learning Outcomes

In this Lecture will cover ..

- To Assess Appropriate Technology: Evaluate the principles and features of appropriate technology, focusing on its ethical design and implementation to address the needs of local communities, especially in developing areas.
- To Apply Ethical Practices: Real-world engineering scenarios, using case studies and practical applications to explore how engineers can promote sustainability, human well-being, and transparency in their work.

CHALLENGER DISASTER AND TECHNOLOGY TRANSFER



The Challenger Disaster: A Case Study



The Challenger Disaster: A Case Study

❖ **Introduction:**

- The space shuttle Challenger disaster remains one of the most poignant reminders of the critical role that ethics play in engineering.
- The disaster, which resulted in the death of seven astronauts, serves as a stark example of what can go wrong when ethical concerns are overridden by other considerations.
- This case study explores the events leading to the disaster, the ethical dilemmas faced by -

The Challenger Disaster: A Case Study

- **Introduction:**
- faced by Roger Boisjoly, and the broader implications for engineering ethics and organizational behavior.
- **Background:**
- Space shuttle Challenger was part of NASA's space shuttle program, designed to provide routine access to space. On January 28, 1986, the shuttle was set to embark on its tenth mission, STS-51-L.

The Challenger Disaster: A Case Study

❖ **Background:**

- Among the crew was Christa McAuliffe, who was to be the first teacher in space, generating significant public interest and pressure to proceed with the launch.
- Roger Boisjoly was an engineer at Morton Thiokol, the contractor responsible for the solid rocket boosters (SRBs) used in the shuttle. He specialized in the seals known as O-rings, which were critical to ensuring the integrity of the SRBs.

The Challenger Disaster: A Case Study

- **Background:**
- **Events Leading to the Disaster**
  - **Identification of the Problem:** Boisjoly and his colleagues had observed that the O-rings became less pliable at low temperatures, which compromised their ability to seal the joints of the Solid Rocket Boosters (SRBs).
  - This was a significant concern because a failure in the O-rings could lead to a catastrophic breach.

The Challenger Disaster: A Case Study

- **Background:**
- **Events Leading to the Disaster**
  - **Internal Warnings:** Boisjoly and other engineers at Morton Thiokol raised their concerns internally.
  - They documented instances where the O-rings showed signs of erosion and blow-by during previous flights, particularly under cold conditions.

The Challenger Disaster: A Case Study

- **Background:**
- **Events Leading to the Disaster**
  - **Pre-Launch Teleconference:** On the evening before the scheduled launch, temperatures at the Kennedy Space Center were predicted to be uncharacteristically low. Boisjoly and his team strongly recommended postponing the launch. During a crucial teleconference with NASA, they presented data showing the risks of launching under such conditions.

The Challenger Disaster: A Case Study

- **Background:**
- **Decision to Launch**
- **Management Override:** Despite the engineers' strong objections, Morton Thiokol's management, under significant pressure from NASA, decided to proceed with the launch.
- The decision was influenced by factors such as schedule pressures, budget constraints, and the high public profile of the mission

The Challenger Disaster: A Case Study

- **Background:**
- **Decision to Launch**
- **Flawed Risk Assessment:** NASA and Morton Thiokol management downplayed the engineers' concerns, arguing that the evidence was inconclusive and that previous flights had been completed without catastrophic failure despite similar conditions.

The Challenger Disaster: A Case Study

The Disaster

- On the morning of January 28, 1986, the Challenger was launched. Just 73 seconds into the flight, the O-rings in the right SRB failed due to the cold temperatures, leading to a breach.
- Hot gases escaped and caused the SRB to separate from the external fuel tank, resulting in the shuttle breaking apart. All seven crew members were killed.

The Challenger Disaster: A Case Study

Ethical Analysis

- **Ethical Responsibility to Public Safety:** Engineers have a fundamental duty to protect public safety. Boisjoly's persistent efforts to delay the launch were driven by this ethical imperative.
- **Integrity and Honesty:** Boisjoly demonstrated professional integrity by openly communicating his concerns, despite facing pressure from management.

The Challenger Disaster: A Case Study

- **Ethical Analysis**
- **Risk Management and Decision-Making:** The Challenger disaster highlights a critical failure in risk management. Ethical decision-making requires a thorough assessment of risks and a commitment to safety, even in the face of external pressures.
- **Organizational Culture:** The culture within NASA and Morton Thiokol at the time discouraged the open expression of concerns.

The Challenger Disaster: A Case Study

- **Ethical Analysis**
- **Organizational Culture:** A healthy organizational culture should encourage dissenting opinions and prioritize safety over other considerations.
- **Accountability:** The disaster prompted a reevaluation of accountability mechanisms within NASA and the aerospace industry. Ensuring accountability at all levels is crucial for preventing similar tragedies.

The Challenger Disaster: A Case Study

- **Consequences and Reforms**
- **Immediate Aftermath:** The disaster led to a suspension of the Space Shuttle program and a thorough investigation by the Rogers Commission. The commission identified the failure of the O-rings and the flawed decision-making processes as key factors.
- **Organizational Reforms:** NASA implemented significant changes to its management and decision-making processes, emphasizing the

The Challenger Disaster: A Case Study

- **Consequences and Reforms**
- **Organizational Reforms:** the importance of safety and encouraging open communication and the consideration of dissenting opinions.
- **Educational Impact:** The Challenger disaster has since become a case study in engineering ethics, risk management, and organizational behavior, taught in engineering and management programs worldwide.

The Challenger Disaster: A Case Study

- **Consequences and Reforms**
- **Public Trust:** The disaster had a profound impact on public trust in NASA. Restoring this trust required significant changes in how NASA conducted its operations and communicated with the public.

The Challenger Disaster: A Case Study

Broader Implications for Engineering Ethics

- **Whistleblowing:** Boisjoly's actions highlight the ethical dilemma of whistleblowing. When internal mechanisms fail to address serious concerns, engineers may face the difficult choice of going public with their warnings.
- **Professional Courage:** The case underscores the importance of professional courage in standing up for ethical principles, even when it may be professionally risky.

The Challenger Disaster: A Case Study

- **Broader Implications for Engineering Ethics**
- **Systemic Change:** The Challenger disaster illustrates the need for systemic changes to prevent ethical lapses. This includes fostering a culture of openness, improving risk management practices, and ensuring that safety is never compromised for other priorities.

The Challenger Disaster: A Case Study

Conclusion

- The Roger Boisjoly and Challenger Disaster case is a powerful reminder of the critical importance of ethics in engineering.
- It underscores the need for engineers to prioritize public safety, maintain integrity, and foster an organizational culture that values ethical decision-making and transparent communication.

The Challenger Disaster: A Case Study

Conclusion

- By learning from such tragedies, the engineering profession can strive to prevent future disasters and ensure that technological advancements serve the greater good.
- The case serves as a lasting testament to the ethical responsibilities of engineers and the profound impact their decisions can have on society.

Ethical Lessons Challenger Disaster

- **Lessons Learned:**
- Ethical Responsibility to Public Safety
- Integrity and Honesty in Engineering
- Risk Management and Decision-Making
- Professional Competence and Due Diligence
- Organizational Culture and Ethics
- Accountability and Responsibility
- Impact on Public Trust

Ethical Considerations- Technology Transfer



Ethical Considerations- Technology Transfer

❖ **Background:**

- Engineering ethics encompasses a set of principles and standards that guide the professional conduct of engineers.
- When considering technology transfer and appropriate technology, several ethical considerations come into play.
- 1. Intellectual Property Rights
- 2. Equity and Fairness 3. Sustainability
- 4. Cultural Sensitivity 5. Capacity Building:

Ethical Considerations- Technology Transfer

- **Intellectual Property Rights:**
- **Respect for IP:** Engineers must ensure that intellectual property rights are respected during technology transfer.
- Unauthorized use or distribution of proprietary technology is unethical and can lead to legal repercussions.
- **Fair Licensing:** Licensing agreements should be fair and transparent, providing mutual benefits to both parties involved.

Ethical Considerations- Technology Transfer

- **Equity and Fairness:**
- **Access to Technology:** Engineers should strive to make advanced technologies accessible to all, including disadvantaged and developing communities, without exploitation.
- **Avoiding Exploitation:** Technology transfer should not exploit less developed regions. Instead, it should empower local communities and contribute to their sustainable development.

Ethical Considerations- Technology Transfer

- **Sustainability:**
- **Environmental Impact:** Engineers must consider the environmental implications of transferring technology. Technologies that cause significant environmental harm should be avoided.
- **Long-term Viability:** The transferred technology should be sustainable in the long term, considering resource availability and environmental impact.

Ethical Considerations- Technology Transfer

- **Cultural Sensitivity:**
- **Cultural Appropriateness:** Technology transfer should be culturally sensitive and compatible with local practices and values. Ignoring cultural contexts can lead to the failure of technology adoption.
- **Capacity Building:**
- **Training and Education:** Engineers have an ethical duty to provide necessary training and education to ensure that the receiving party -

Ethical Considerations- Technology Transfer

- **Capacity Building:**
- **Training and Education:** receiving party can effectively use and maintain the transferred technology.
- **Empowerment:** Technology transfer should aim to build local capacity and empower communities to become self-sufficient.

Ethical Considerations- Technology Transfer

- **Human Well-being:**
- **Health and Safety:** Appropriate technology should prioritize the health and safety of users, reducing risks and improving living conditions.
- **Quality of Life:** The technology should enhance the quality of life, addressing basic needs and improving daily living.

Ethical Considerations- Technology Transfer

- **Sustainability and Environmental Stewardship:**
- **Environmental Responsibility:** Appropriate technology should minimize environmental impact and promote the use of renewable resources.
- **Resource Efficiency:** Engineers should design technologies that are resource-efficient and reduce waste.

Ethical Considerations- Technology Transfer

- **Affordability and Accessibility:**
- **Cost-effectiveness:** Appropriate technology should be affordable for the target community, ensuring broad accessibility.
- **Inclusivity:** The technology should be inclusive, considering the needs of all community members, including marginalized groups.

Ethical Considerations- Technology Transfer

- **Ethical Design and Innovation:**
- **User-Centric Design:** Engineers should involve local communities in the design process to ensure that the technology meets their specific needs and preferences.
- **Innovation for Good:** Innovations should aim to solve real problems and provide tangible benefits to the community.

Ethical Considerations- Technology Transfer

- **Transparency and Accountability:**
- **Honest Communication:** Engineers should communicate openly and honestly about the capabilities and limitations of the technology.
- **Accountability:** Engineers should take responsibility for the outcomes of their work, including any unintended negative consequences.

Examples

1) **Technology Transfer in Agricultural Engineering:**

- **Green Revolution:** The Green Revolution involved the transfer of high-yield crop varieties and modern agricultural techniques to developing countries.
- While it significantly increased food production, it also raised ethical concerns about environmental impact, social inequality, and the displacement of traditional farming practices.

Examples

2) **Appropriate Technology in Water Management:**

- **Biosand Filters:** Biosand filters are an example of appropriate technology used to provide clean drinking water in rural communities.
- They are affordable, easy to maintain, and effective in removing contaminants, thereby improving health and reducing waterborne diseases.

- 3. Renewable Energy Projects:

Examples

3) **Renewable Energy Projects:**

- **Solar Panels in Remote Areas:** Installing solar panels in remote areas provides a sustainable and renewable source of energy, enhancing energy access and reducing reliance on fossil fuels.
- Ethical considerations include ensuring affordability, local training for maintenance, and respecting local environmental conditions.

Thanks

Any Question

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